

“borrowed” dollars that can be in the system relative to the “real” supply. For instance, there is occasionally more DAI in Compound than is actually supplied by MakerDAO because users are borrowing and resupplying or selling to others who resupply. Importantly, all MakerDAO supply is ultimately backed by real collateral, and there is no way to borrow more collateral value than has been supplied.

For example, suppose an investor deposits 100 DAI with a collateral factor of 90. This transaction alone corresponds to a required collateralization ratio of 111 percent. Assuming 1 DAI = \$1, the investor can borrow up to \$90 worth of any other asset in Compound. If he or she borrows the maximum and the price of the borrowed asset increases at all, the position is subject to liquidation. Suppose the investor also deposit two ETH with a collateral factor of 60 and a price of \$200/ETH. The total supply balance is now \$500, with 80 percent being ETH and 20 percent being DAI. The required collateralization ratio is  $100/(0.8*60 + 0.2*90) = 151$  percent (Figure 6.2).

The supply and borrow interest rates are compounded every block (approximately 15 seconds on Ethereum producing near continuous compounding) and are determined by the utilization percentage in the market. Utilization is calculated as total borrow/total supply. The utilization rate is used as an input parameter to a formula that determines the interest rates. The remaining parameters are set by *Compound Governance*, which we describe near the end of this section.